Varun Satish

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Education

University of Illinois Urbana-Champaign August 2022 - May 2026
Bachelor of Science, Astrophysics + Data Science with Distinction
Award: Distinguished Undergraduate Researcher Certificate (May 2025)

Work Experience

Department of Astronomy, University of Illinois Urbana-Champaign July 2026 – Present
Post Baccalaureate Researcher

Department of Astronomy, University of Illinois Urbana-Champaign August 2023 – May 2026
Telescope Observing Teaching Assistant

- Trained 30+ students on the operation and preservation of a national historic observatory telescope.
- Instructed new students on foundational astronomical concepts and observational techniques.
- Led 8 semesters of night observing and solar observing sessions for students in introductory astronomy courses, facilitating hands-on learning experiences.

Department of Astronomy, University of Illinois Urbana-Champaign September 2023 – May 2026
Course Grading Assistant

- Graded and provided detailed feedback on discussion assignments for ASTR 330: Extraterrestrial Life (Prof. Dunne), promoting deeper engagement with astrobiology topics.
- Supported grading and instructional activities for ASTR 100: Introduction to Astronomy, ASTR 210: Introduction to Astrophysics, and ASTR 310: Computing in Astronomy, ASTR 406: Galaxies and Cosmology.

Campus Leadership Experience

Astronomical Society, University of Illinois Urbana-Champaign April 2024 – May 2026
President

- Led planning and execution of Observatory Open Houses, new student events, and community star parties, fostering an inclusive and active astronomy community.
- Partnered with the Department of Astronomy to engage prospective students through admissions events, orientations, and outreach initiatives.
- Designed and ran outreach events for high schools, university students, and the general public.
- Worked to improve engagement in public events by introducing new exhibits, fun displays, and improving social media content and outreach.

Friends of the University of Illinois Observatory April 2024 – May 2026
Board Member

- Appointed as the Astronomical Society representative to advance preservation goals of the historic Campus Observatory.
- Initiated programs to strengthen alumni connections and increase community participation in observatory events and heritage efforts.

Illinois Space Society, University of Illinois Urbana-Champaign October 2022 – May 2024
Educational Outreach Board Member

- Supported STEM outreach initiatives connecting local schools with space and astronomy education opportunities
- Coordinated interactive events and workshops for middle and high school students to promote engagement in science and engineering.
- Developed and delivered educational content designed to inspire early interest in space exploration and STEM careers.

Research Experience

Astronomy Education Research, University of Illinois Urbana-Champaign January 2025 – Present
Undergraduate Researcher

- Investigated the extent and quality of astronomy education in rural Illinois schools, compared to the Next Generation Science Standards (NGSS).
- Employed qualitative analysis software to code and interpret semi-structured teacher interviews across the state.
- Co-authored two research papers to disseminate findings and develop outreach resources that support teacher programming.

Cosmology/Radiative Transfer Research, University of Illinois August 2023 – Present
Undergraduate Researcher

- Conducted research on galaxy evolution and the propagation of radiation from stars in early galactic environments, contributing to a deeper understanding of radiative transfer.
- Utilized Python packages including Numpy, Healpy (HEALPix for Python), yt, and matplotlib, along with parallel computing techniques to process and analyze large simulations.
- Currently investigating ionizing radiation attenuation across galaxies by calculating the Escape Fraction of Ionizing Radiation, a key metric in understanding radiation's journey from stars through the interstellar medium.

Publications

Published

- Nguyễn, T. H., Barrow, K. S. S., Byrom, L., **Satish, V.** (2025). "Effect of molecular hydrogen self-shielding modeling on early Reionization Era galaxies in radiative hydrodynamic cosmological simulations." *MNRAS*. https://doi.org/10.1093/mnras/staf2045

Submitted

- Wilken, S., Looney, L. W., Talafian, H., **Satish, V**. "What gets Taught? An exploration of astronomy content and curriculum development in rural high schools (Submitted: APS Physical Review PER).
- Barrow, K. S. S., et al. (including **Satish, V.**) "The AGORA High-resolution Galaxy Simulations Comparison Project. XI..." (Submitted: The Astrophysical Journal). https://doi.org/10.48550/arXiv.2605.24097
- **Satish, V.,** Barrow, K. S. S., Byrom, L., Dunham, E. " Exploring Escape Fractions in ENZO Simulations of the Epoch of Reionization" (Submitted: The Astrophysical Journal).
- Wilken, S., Looney, L. W., **Satish, V**., Hussey, M. "Changing Opportunities to Enroll in Astronomy and Physics Courses in U.S. Public High Schools" (Submitted: APS Physical Review PER).

In Prep

- Wilken, S., Looney, L. W., Talafian, H., **Satish, V**. " Motivations Behind Astronomy Instruction in Rural High Schools (In Prep: APS Physical Review PER).
- **Satish, V**., Barrow, K. S. S., Byrom, L. "Implementing a Redshift-Dependent IMF in ENZO Simulations" (In Prep: The Astrophysical Journal)

Conference Participation

247th Meeting of the American Astronomical Society January 2026
American Astronomical Society, Phoenix, AZ

Undergraduate Research Symposium April 2025
University of Illinois Urbana-Champaign

Astrofest 2026 April 2026
University of Illinois Urbana-Champaign

Astrofest 2025 April 2025
University of Illinois Urbana-Champaign